

Pulse Mint White Paper

A 0%-interest, multi-collateral stablecoin protocol for PulseChain

Draft v0.2, 6 November 2026 (adds the price-feed safety rules in section 10). Testnet preview, not audited.

1. Overview

Pulse Mint is a decentralised borrowing protocol for PulseChain. Users lock one of seven supported tokens in a **vault** and mint **USDPM**, a stablecoin designed to track US$1.

Borrowing carries **0% interest**. The only cost is a small **one-off borrowing fee (0.5%)** added to the debt when USDPM is minted. A second token, **PMNT**, has a fixed supply. Staking it earns a share of protocol fees, and it is used to reward Stability Pool depositors.

The design follows the mechanics of Liquity v1 (MIT-licensed). It is re-implemented from scratch and extended to many collaterals, so each collateral runs in its own isolated **branch**. No Liquity v2 (BUSL-licensed) code is used.

Status:

- Fully deployed on the **PulseChain testnet v4 (chain 943)** with mock tokens and mock price feeds.
- Not audited.
- No mainnet contracts exist.

2. The problem

PulseChain holders who need dollar liquidity usually have to **sell** their assets and give up their exposure. The alternative is borrowing on platforms with **variable interest**, which grows the debt over time. A stablecoin that is:

- **over-collateralised**,
- **immutable**,
- free of interest rates set by governance, and
- redeemable at face value

offers a simpler, more predictable way to borrow. Supporting the ecosystem's main tokens in separate branches lets one token's risk stay contained.

3. How vaults work

A vault holds one user's collateral and debt for one branch.

Rule	Value (from the contracts and deploy script)
Interest	0%
Borrowing fee	0.5% of USDPM drawn, one-off (`borrowFeeBps = 50`)
Liquidation reserve	200 USDPM added to the debt, refunded on close (`gasComp = 200e18`)
Minimum debt	2,000 USDPM including the reserve (`minDebt = 2000e18`)
Minimum collateral ratio (MCR)	Per collateral, 120%–200% (see section 9)
Debt cap	Per collateral (see section 9)

Operations:

- **Open:** deposit collateral and choose the USDPM amount.
- Debt = amount + fee + reserve.
- The vault's ratio must be at least the MCR, and the system must stay out of recovery mode.
- **Adjust:** add or withdraw collateral, borrow more, or repay. Each change is re-checked against the same rules.
- **Close:** repay the debt minus the reserve and receive all collateral back.
- **Ordering:** vaults are kept on-chain in a list sorted by *nominal* collateral ratio, which doesn't depend on price. That way the riskiest vault can always be found cheaply.

4. USDPM and the peg

USDPM is minted only by branches and only against collateral. Every USDPM in circulation equals vault debt, a property checked by invariant tests. The peg holds through two mechanisms:

1. **Floor at about $1 (redemptions).** Anyone can hand in USDPM and receive **$1 of collateral per USDPM, minus a 0.5% redemption fee** (`redemptionFeeBps = 50`).
2. Redemptions take collateral from the **lowest-ratio vault first**, skipping vaults below MCR.

3. If USDPM trades under $1, redeeming it returns more than it costs to buy, and that buying pressure lifts the price.
4. A redeemed borrower loses no value: their debt falls by the same dollar amount as their collateral.
5. **Ceiling (over-collateralised minting).** If USDPM trades above $1, borrowers have a reason to mint and sell it. The 0.5% fee marks roughly the band above which minting becomes attractive.

5. 0% interest and the one-off fee

There is no ongoing interest. Debt never grows on its own.

- The **0.5% one-off borrowing fee** is charged in USDPM and sent to PMNT stakers.
- The **0.5% redemption fee** is charged in collateral and also goes to PMNT stakers.
- Both fees are fixed in code (`immutable`). Nobody can change them after deployment.

6. PMNT tokenomics, staking and issuance

Item	Value
Total supply	**100,000,000 PMNT**, minted once at deployment. There is no mint function.
Stability Pool rewards	**32,000,000 PMNT (32%)** held by `CommunityIssuance`
Remaining	68,000,000 PMNT, held by the deployer on testnet. The deploy script earmarks 3% for LP rewards and 65% for treasury/team, as placeholders. The final mainnet allocation and vesting are **not yet decided**.

Issuance streams PMNT to each branch's Stability Pool at a fixed rate per second. Testnet rates:

Branch	PMNT per second
WPLS	0.10
HEX	0.08
PLSX	0.06
INC	0.03
pDAI	0.03
eHEX	0.05
pWETH	0.05
Total	**0.40**

At these rates the 32M pool would last about 80,000,000 seconds, roughly 2.5 years. Mainnet rates are not yet set.

Staking: PMNT stakers receive all borrowing fees (in USDPM) and redemption fees (in each collateral), in proportion to their stake. There is no lock-up. Fee income depends entirely on usage and is not guaranteed.

LP rewards: an `LPRewards` contract can stream PMNT to liquidity providers at a fixed rate for a fixed duration. It is not yet deployed or funded.

7. The Stability Pool and liquidations

A vault whose ratio falls **below its MCR** can be liquidated by anyone.

1. **Liquidator reward: 0.5% of the collateral** (`LIQUIDATOR_BPS = 50`) plus the **200 USDPM reserve**.
2. **Stability Pool offset:** USDPM in the branch's Stability Pool cancels the vault's debt. Depositors receive the rest of the collateral in proportion to their share. Because liquidations happen below the MCR (120%–200%), this collateral is normally worth more than the USDPM used.
3. **Redistribution:** any debt the pool can't cover is shared, with its collateral, among the remaining vaults in proportion to their collateral.

Depositors also earn PMNT issuance. Withdrawals from the pool pause while a liquidatable vault exists, so depositors can't dodge pending losses. Claiming gains still works.

8. Recovery mode

Each branch has a **critical collateral ratio (CCR)**. If the branch's total ratio (TCR) falls below it, the branch enters recovery mode:

- no collateral withdrawals;
- vaults can't be closed;
- new vaults, and any extra borrowing, need a ratio of at least the CCR;
- repaying and adding collateral are always allowed.

Unlike Liquity v1, Pulse Mint does **not** liquidate vaults above MCR in recovery mode. It relies on these rules to protect the system instead.

9. The seven collaterals

Collateral	MCR	CCR	Testnet debt cap (USDPM)
WPLS	150%	200%	20,000,000
HEX	175%	250%	10,000,000
PLSX	150%	200%	10,000,000
INC	200%	300%	500,000
pDAI	150%	200%	5,000,000
eHEX	175%	250%	1,000,000
pWETH	120%	150%	20,000,000

Each branch is isolated. A failure in one collateral can't drain another branch's collateral. All parameters are `immutable`. The mainnet token addresses were checked against several

independent sources: on-chain reads, PulseScan, CoinGecko, the Piteas list, PulseX pairs, and the OmniBridge mapping.

10. Prices (oracles) and safety pauses

Pulse Mint needs to know what each collateral is worth in US dollars. This price decides how much you can borrow, when a vault can be liquidated, and how much collateral a redemption pays out. If the price is wrong, people can lose money unfairly, so the price system follows a few simple rules. They were designed after studying the price-feed problems Liquid Loans had on PulseChain.

10.1 Where prices come from

- **Real markets, averaged.** Prices come from real PulseX trading pools. Instead of the price at one instant, Pulse Mint uses a **30-minute average** (called a TWAP, a "time-weighted average price").
- **Why an average?** One big trade can push a pool's price for a moment. An average over 30 minutes makes that trick very expensive and mostly useless, so nobody can briefly fake a price to liquidate you or borrow against inflated collateral.
- **The trade-off:** in a very fast market, the app's price can lag what you see on PulseX by a few minutes. That is expected, not a bug. Keep a healthy collateral ratio so a quick drop doesn't catch you out.
- **Pools must have enough money in them.** A price only counts if its pool still holds enough funds (roughly half of its size when the setting was chosen). If a pool is drained and becomes cheap to push around, its price is ignored.

10.2 Two price sources, with a different backup

Every collateral is checked against **two** sources:

- **Main price:** the coin's PulseX average price against PLS, multiplied by PLS's dollar price from a deep PLS/USDC PulseX pool.
- **Backup price:** a **different** PulseX pool for the coin, multiplied by PLS's dollar price from the **Fetch Oracle**. Fetch is a separate price service run by reporters, not a PulseX pool, so a problem on PulseX doesn't knock out both sources at once. Pulse Mint only uses Fetch prices that are at least 20 minutes old (so wrong reports can be challenged first) and no more than 4 hours old, and it never holds on to a report that has been challenged and removed.

Honest limit: Fetch only reports PLS's dollar price today. For the other coins, the coin-to-PLS part of the backup still comes from PulseX, just from a different pool.

10.3 pDAI is not $1

pDAI is the copy of DAI that was created when PulseChain forked from Ethereum (token `0x6B175474E89094C44Da98b954EedeAC495271d0F`). It is not backed like real DAI and its price moves up and down; it was around **$0.00114** in November 2026. Pulse Mint prices pDAI from its live PulseX market like every other coin, and **never treats it as $1**. Treating it as $1 would let people borrow far more than their collateral is worth.

10.4 What happens when a price is too old

Sometimes both price sources can stop giving a good price, for example if PulseX has trouble. Pulse Mint then keeps using the last good price, but only for a limited time, and it pauses only the actions that would be risky:

How long without a fresh price	What pauses	What always keeps working
A few minutes (a short glitch)	**Nothing**	Everything
More than 15 minutes	**Redemptions**	Borrowing, repaying, adding and withdrawing collateral, closing, liquidations
More than 2 hours	Redemptions, **new borrowing** and **withdrawing collateral**	Repaying, adding collateral, closing a loan, liquidations

Everything reopens by itself as soon as a fresh price arrives. Nobody has to switch it back on.

10.5 What happens when the two sources disagree

If the main and backup prices are more than 5% apart, something may be wrong with one of them. Then:

- **redemptions pause**, so nobody can swap USDPM for collateral at a possibly wrong rate;
- the **lower** of the two prices is used, which is the safer choice for the system;
- **liquidations keep running**, so unsafe vaults can still be cleared.

When the sources agree again, redemptions reopen automatically. The app shows a yellow notice on the Borrow, Redemptions and Liquidations pages whenever any pause is on, and says why.

10.6 Why these rules are good for users

- **No borrowing against a fake or stale price.** If the price is hours old, nobody can take out new USDPM or pull collateral out based on a value that may no longer be true. That protects every USDPM holder.
- **Fewer unfair redemptions.** Redemptions pause when prices are unsettled, so nobody can grab cheap collateral from your vault using a wrong price.
- **Safer in a crash.** Liquidations never pause, so the system can still clear underwater vaults and USDPM stays backed when prices fall.
- **Your own safety moves are never blocked.** You can always repay, add collateral or close your loan, so you can protect your vault even during a pause.
- **Everyday use isn't constantly interrupted.** Short glitches pause nothing, and everything reopens on its own.
- **A fix for a known weak spot.** Liquid Loans' audit found that if both of its price sources failed, it could keep using the last good price forever. Pulse Mint puts a time limit on this.

10.7 Testnet vs mainnet

The testnet uses test price feeds whose prices the team sets by hand, so the app can be tried out. These feeds support the same pause signals, so the notices can be demonstrated. The real price

setup above is planned for mainnet and will be checked again just before any launch.

11. Security and audit plan

- **Tests:** 61 Foundry tests, all passing. They cover unit tests, fuzz tests (vault ops, liquidations, redemptions, the Stability Pool, recovery mode) and invariant tests that run random sequences of actions. The invariants check that:
- USDPM supply equals system debt;
- collateral bookkeeping balances;
- the sorted list stays in order;
- the Stability Pool stays solvent.
- **Static analysis:** Slither found no high-severity issues.
- **Internal review:** two bugs were fixed (see `docs/security-review.md`).
- **Open item:** on testnet the deployer wallet still controls the wiring contracts (USDPM, PMNTStaking, CommunityIssuance). Before mainnet this must be renounced, or moved to a multisig behind a timelock.
- **Plan:**
- External audit or audits.
- Fix and re-review.
- Public bug bounty.
- Capped mainnet.

12. Roadmap

1. **Testnet (now):** all seven branches live on PulseChain testnet v4, with the web app and analytics.
2. **Audit:** professional external review, then a public report and a bug bounty.
3. **Capped mainnet:** launch with low debt caps and locked-down admin roles, then raise caps gradually.

There are no dates yet. Each stage starts only after the previous one is complete.

13. Risks

- **Smart-contract risk:** bugs may exist despite testing. The code is unaudited.
- **Liquidation risk:** sharp price drops can liquidate vaults. Borrowers then lose their collateral but keep the USDPM they borrowed.
- **Oracle risk:** thin liquidity, manipulation or outages could produce wrong or stale prices. Section 10 explains the limits and pauses that reduce this risk; they cannot remove it completely.
- **Peg risk:** USDPM may trade away from $1, especially in stressed markets.
- **Collateral risk:** the value of a collateral token may collapse. Bridged tokens (pDAI, eHEX, pWETH) also depend on the bridge.
- **Stability Pool risk:** deposits shrink when used, and collateral gains may be worth less than expected if prices keep falling.
- **Admin-key risk** (until renounced): see section 11.

- **Regulatory risk:** rules on stablecoins and DeFi may change.

14. Disclaimer

This document describes software under development. It is not an offer, solicitation or financial advice, and nothing in it is a promise of returns or of a launch date. Parameters may change before any mainnet deployment. Use of the software is at your own risk. **Testnet preview, coming soon. Not financial advice. Crypto is risky.**